

Relation is a connection between objects/elements.

e.g. Father-Son relation, Mother-Daughter Relation.

e.g. Ram is father of Ravi.

Father (F) = {(Ram, Ravi), (Surya, Vrinda), (Raju, Pawan)}

Here, Ram and Ravi are the objects related by relation Father.

e.g. 4 is greater than 3

Here, 4 and 3 are objects related by relation greater than.

Let A and B be sets. A binary relation from A to B is a subset of $A \times B$.

In other words, a binary relation from A to B is a set R of ordered pairs where the first element of each ordered pair comes from A and the second element comes from B .

Example:

$$A = \{1, 2, 3\} \text{ and } B = \{3, 5\}$$

$$A \times B = \{(1, 3), (1, 5), (2, 3), (2, 5), (3, 3), (3, 5)\}$$

$$R = \{a R b \mid a < b (a \in A, b \in B)\}$$

$$R = \{(1, 3), (1, 5), (2, 3), (2, 5), (3, 5)\}$$

$$R \subseteq A \times B$$

$$\text{Domain}(R) = \{a \mid a \in A \text{ and } (a, b) \in R \text{ for some } b \in B\}$$

$$\text{Range}(R) = \{b \mid b \in B \text{ and } (a, b) \in R \text{ for some } a \in A\}$$

$$a R b \iff (a, b) \in R$$

$$a \not R b \iff (a, b) \notin R$$

A relation on the set A is a relation from A to A .

Example: $A = \text{set } \{1, 2, 3, 4\}$.

Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$

Solution: Since (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b

$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}$

How many relations are there on a set with n elements?

A relation on a set A is a subset of $A \times A$. Because $A \times A$ has n^2 elements when A has n elements and a set with m elements has 2^m subsets, there are 2^{n^2} subsets of $A \times A$.

Thus there are 2^{n^2} relations on a set with n elements.

For example there are $2^{3^2} = 2^9 = 512$ relations on the set $\{a, b, c\}$

- ▶ There are many relations from set A to A
- ▶ Each relation on $A \subseteq A \times A$
- ▶ $\{\}$ denotes the void relation
- ▶ $A \times A$ denotes the Universal Relation

Operations on relations

Since a relation is a set of ordered pairs, two relations R and S from X to Y can be combined.

So the following operations can be performed on Relations, just like Sets.

1. Union
2. Intersection
3. Difference
4. Complement

Example

Let $A = \{1, 2, 3, 4\}$

$R = \{(1, 1), (2, 3), (1, 4), (3, 4)\}$ and

$S = \{(1, 1), (2, 2), (2, 3), (3, 1), (4, 3)\}$ are relations on set A .

Find $R \cup S$, $R \cap S$, $R - S$ and $\overline{R}$

$$R \cup S = \{(1, 1), (2, 3), (1, 4), (3, 4), (2, 2), (3, 1), (4, 3)\}$$

$$R \cap S = \{(1, 1), (2, 3)\}$$

$$R - S = \{(1, 4), (3, 4)\}$$

$$\overline{R} = (A \times A) - R$$

Composition

Let R be a relation from X to Y and S be a relation from Y to Z . Then composition of R and S denoted as RoS is a relation from X to Z defined as

$$RoS = \{(x, z) \mid \text{there exists } y \in Y \text{ such that } (x, y) \in R \text{ and } (y, z) \in S\}$$

The composition of relation R with itself is $RoR \implies R^2$
 $R^2 o R \implies R^3$

Let $A = \{1, 2, 3, 4\}$

$R = \{(1, 3), (2, 4), (1, 2)\}$ and $S = \{(3, 1), (4, 3), (2, 3), (4, 1)\}$ are relations on set A .

Find RoS and SoR

$RoS = \{(1, 1), (2, 3), (2, 1), (1, 3)\}$

$SoR = \{(3, 3), (3, 2), (4, 2), (4, 3)\}$

$RoS \neq SoR$ (Composition of Relation is not Commutative)

Let R be a relation from a set X to set Y . The inverse of the relation R denoted by R^{-1} is the relation from Y to X and is defined as:

$$R^{-1} = \{(y, x) : (x, y) \in R\}$$

e.g. Let $R = \{(1, 3), (2, 5), (4, 7)\}$ be a relation on a set $A = \{1, 2, 3, 4, 5, 6, 7\}$. Find R^{-1}

$$R^{-1} = \{(3, 1), (5, 2), (7, 4)\}$$

Representation of Relations

► Listing

Let $A = \{1, 2, 3\}$

e.g. $R = \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

► Statement

e.g. Set of all ordered pairs where first element is not more than second

► Set Builder

e.g. $R = \{(x, y) | x \leq y\}$

► Matrix (1-Relation, 0-No Relation)

Let R be a relation from $X = \{a_1, a_2, \dots, a_n\}$ to $Y = \{b_1, b_2, \dots, b_m\}$.

Then $M_R = [a_{i,j}]_{n \times m}$, where

$$a_{i,j} = \begin{cases} 1 & \text{if } a_i R a_j \\ 0 & \text{if } a_i \not R a_j \end{cases} \quad (1)$$

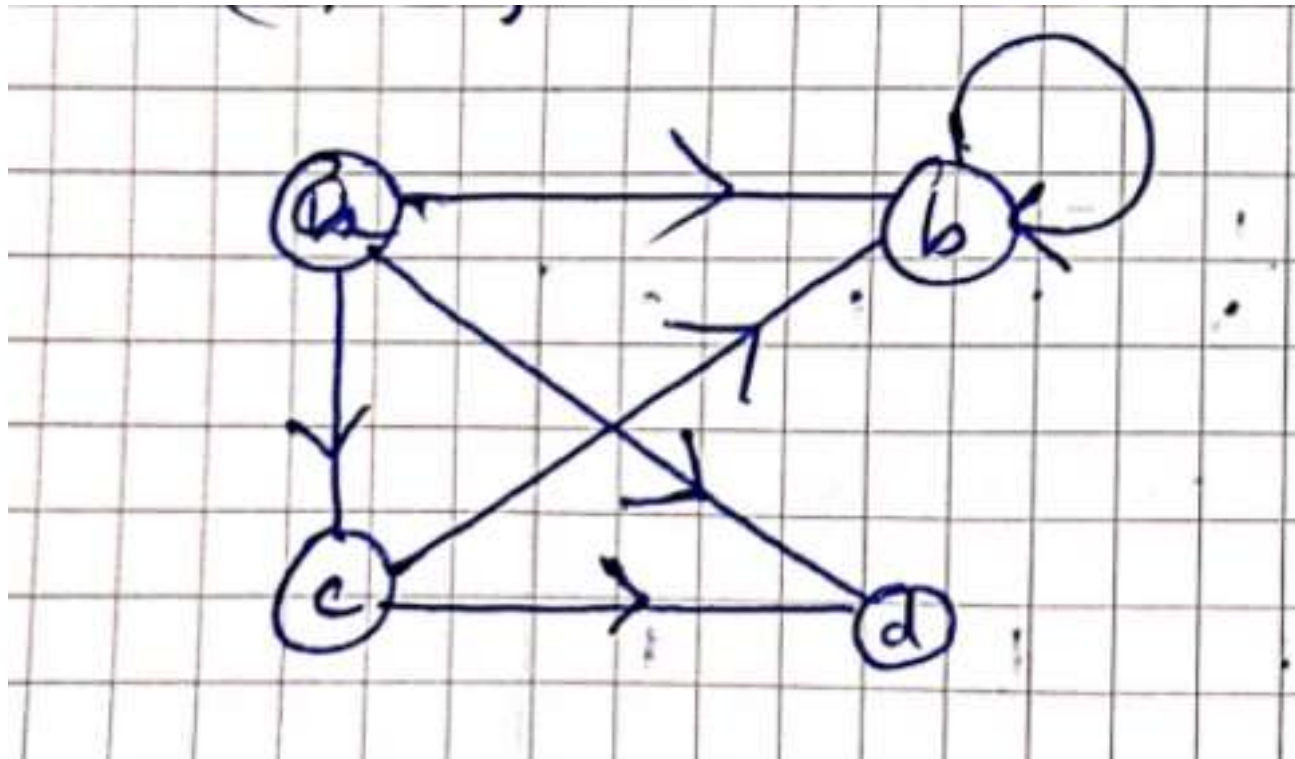
Example

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 3), (3, 2), (3, 4), (1, 2), (4, 1)\}$ be relation defined on A . Therefore, the matrix of this relation R is given by-

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

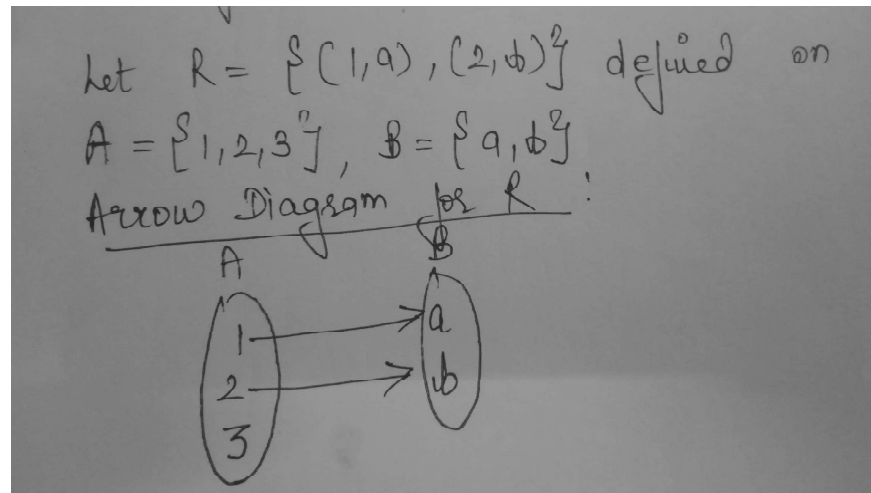
Representation of Relations

- Digraph(only for Relation $A \Rightarrow A$)
Objects are represented as Nodes
Relation is represented by line with arrows
Let $X = \{a, b, c, d\}$
and $R = \{(a, b), (a, c), (c, b), (a, d), (b, b), (c, d)\}$ be a Relation defined on X .

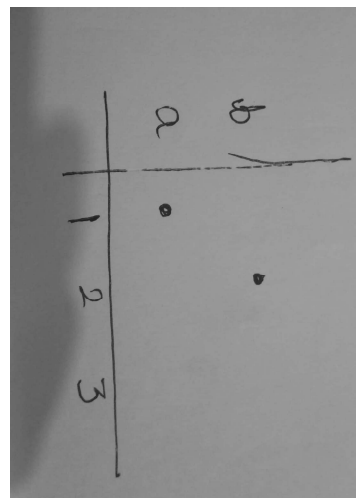


Representation of Relations

► Arrow Diagram



► Graph



► Table

Let $V = \{1, 2, 3, 4\}$ and $I = \{0.1, 0.2, 0.3, 0.4\}$

Relation $R = \{(1, 0.1), (2, 0.2), (3, 0.3), (4, 0.4)\}$ can be represented as

V	I
1	0.1
2	0.2
3	0.3
4	0.4

1. Reflexive Relation

Definition: Let R be a relation defined on a set X . The Relation R is said to be Reflexive if $x R x$ or $(x, x) \in R$ for all $x \in X$.

e.g. $M = \{1, 2, 3\}$

$(R, \leq) = \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

$\forall x \in M, (x, x) \in R$

In terms of Digraph, all self loops must be present.

In terms of Matrix, all diagonal elements must be one.

Let $A = \{1, 2, 3, 4\}$. Determine whether the following Relations are Reflexive.

1. $R_1 = \{(1, 1), (3, 3), (2, 4), (4, 4)\}$
2. $R_2 = \{((1, 1), (2, 2), (2, 4), (3, 3), (4, 4)))\}$

2. Irreflexive Relation

Definition: The Relation R is said to be Irreflexive if $\forall x \in A, x \not R x$ or $(x, x) \notin R$.

e.g. $M = \{1, 2, 3\}$

$R = \{(1, 2), (2, 1), (2, 3)\}$

$\forall x (x \in M \implies (x, x) \notin R)$

In terms of Digraph, all self loops must not be present.

In terms of Matrix, all diagonal elements must be zero.

Check whether the following Relation are irreflexive

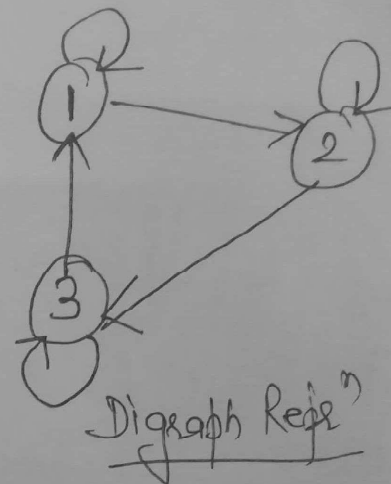
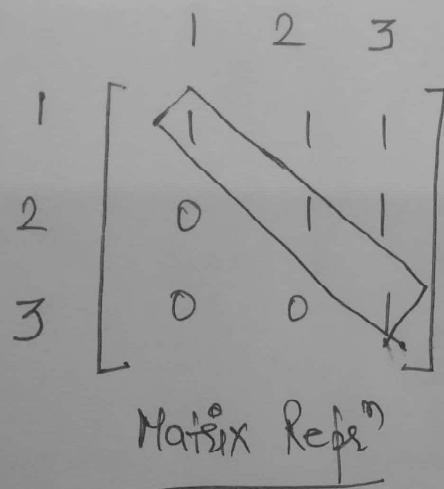
1. $R = \{(1, 2), (2, 3), (1, 3)\}$

2. $R = \{(1, 1), (2, 2), (1, 3)\}$

If a Relation is not Reflexive, it does not indicate it is Irreflexive. It can be none.

Reflexive Relations($=, <=, >=$)

Irreflexive Relations ($<, >$).



→ 4 Diagonal elements are all 1, then Reflexive
 all 0, then Irreflexive
 else None

→ 4 all nodes have self loops, then Reflexive
 No self loops, then Irreflexive
 else None.

3. Symmetric Relation

Definition: Let R be a Relation defined on a set X . The relation R is said to be Symmetric if $x R y$, then $y R x$, for all $x, y \in X$

$$(x, y) \in R \implies (y, x) \in R \text{ for all } x, y \in X$$

e.g. $S = \{1, 2, 3\}$

$$R = \{(1, 2), (2, 1), (3, 3)\}$$

Check for $R^2 = \{(1, 2), (2, 1), (1, 3)\}$

(Not Symmetric)

4. Anti-Symmetric Relation

Definition: Let R be a Relation defined on a Set X , The Relation R is said to be Anti-Symmetric if $x R y$ and $y R x$, then $x=y$ for all $x, y \in X$ or

$(x, y) \in R$ and $(y, x) \in R \implies x = y$ for all $x, y \in X$ e.g

$R = \{(1,2), (1,4), (3,3)\}$

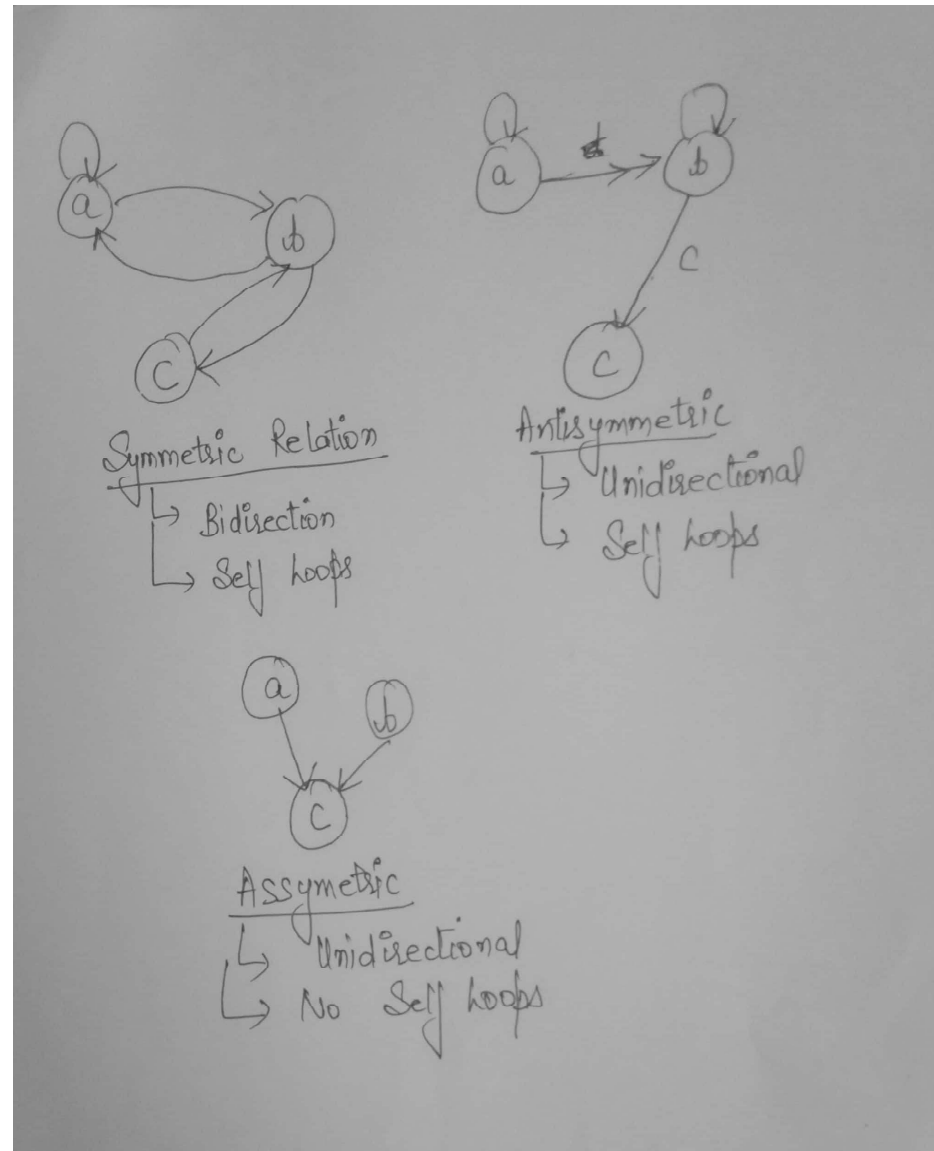
In terms of Digraph, self loops are allowed.

5. Asymmetric Relation

Definition: The Relation R is said to be Assymetric if $x R y$, then $y \not R x$
or

$$(x, y) \in R \implies (y, x) \notin R \text{ for all } x, y \in X$$

In terms of Digraph, self loops are not allowed.



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Possible to have a Relation which is both Symmetric and Antisymmetric

e.g. $S = \{1, 2, 3, 4\}$

$R = \{(1, 1), (2, 2), (3, 3)\}$

Every Asymmetric Relation is also Antisymmetric

Asymmetric $\subseteq$ *Antisymmetric*

6. Transitive Relation

Definition: Let R be a Relation defined on a Set X . The Relation R is said to be Transitive if $x R y$ and $y R z$ then $x R z$ for all $x, y, z \in X$.

Or

$(x, y) \in R$ and $(y, z) \in R \implies (x, z) \in R$ for all $x, y, z \in X$

e.g. $S = \{1, 2, 3, 4\}$

$R = \{(1, 2), (2, 3), (1, 3), (3, 4), (1, 4), (2, 4)\}$ is Transitive in nature.

Operations that are Transitive:-

$=, <=, >=, \subseteq$

Let R be a relation on a set of positive integers defined as $x R y$ if and only if $x|y$ i.e x divides y . Determine whether the relation R is transitive or not.

$x|y$ implies that $y = k_1x$ ($k_1 \in Z$) and $y|z$ implies that $z = k_2y$ ($k_2 \in Z$).

Thus $z = k_2k_1x$ which implies $x|z$

Since $x R y$ and $y R z$ implies $x R z$, the Relation R is Transitive.

Let R be a relation defined on set X .

The Relation R is said to be **compatible relation** if it is reflexive and symmetric.

The Relation R is said to be an **equivalence relation** if it is Reflexive, Symmetric and Transitive.

The relation R is said to be a **partial order relation** if it is reflexive, anti-symmetric and transitive.

1. Reflexive Closure:-

The Reflexive Closure of a Relation R defined on a non-empty set X is the smallest reflexive relation that contains R as a subset.

Let I_X be the Identity Relation defined on a Set X .

$$I_X = \{(a, a) : a \in X\}$$

Then the Reflexive Closure of R is

$$R \cup I_X$$

Example

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 3), (2, 2), (1, 4)\}$

Find the Reflexive Closure of R .

$$I_A = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R \cup I_A = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 3), (1, 4)\}$$

2. Symmetric Closure:-

The Symmetric Closure of a Relation R defined on a non-empty set X is the smallest Symmetric relation that contains R as a subset.

It is the set $R \cup R^{-1}$ Let $A = \{1, 2, 3, 4\}$ and

$R = \{(1, 3), (2, 2), (1, 4), (3, 4)\}$

Find the Symmetric Closure of R .

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 3), (2, 2), (1, 4), (3, 4)\}$

Find the Symmetric Closure of R .

$$R^{-1} = \{(3, 1), (2, 2), (4, 1), (4, 3)\}$$

$$R \cup R^{-1} = \{(1, 3), (2, 2), (1, 4), (3, 1), (3, 4), (4, 1), (4, 3)\}$$

3. **Transitive Closure**:- The Transitive Closure of a Relation R defined on a non-empty set X is the smallest Transitive relation that contains R as a subset.

It is denoted by R^∞

$$R^\infty = \bigcup_{i=1}^{\infty} R_i$$

In general, if the set X has n elements, then

$$R^\infty = R \cup R^2 \cup R^3 \dots R^n$$

Example

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 3), (2, 2), (2, 4), (3, 4)\}$

Find the Transitive Closure of R .

$$R^2 = R \circ R = \{(1, 4), (2, 2), (2, 4)\}$$

$$R^3 = R^2 \circ R = \{(2, 2), (2, 4)\}$$

$$R^4 = R^3 \circ R = \{(2, 2), (2, 4)\}$$

$$\begin{aligned} \text{Therefore, } R^\infty &= R \cup R^2 \cup R^3 \cup R^4 \\ &= \{(1, 3), (2, 2), (2, 4), (3, 4), (1, 4)\} \end{aligned}$$

Equivalence Relation and Partition

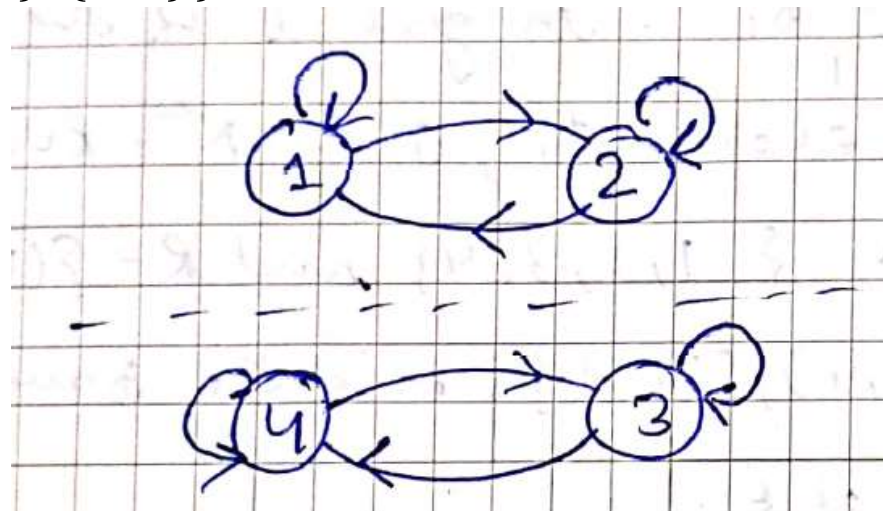
Let $A = \{1, 2, 3, 4\}$

$R = \{(1,1), (1,2), (2,1), (2,2), (3,3), (3,4), (4,3), (4,4)\}$

The given relation is Reflexive, Symmetric and Transitive

Therefore, it is an equivalence relation.

Partition $P = \{\{1, 2\}, \{3, 4\}\}$



A Partition of a set X is a set of non-empty subsets of X such that every element x in X is in exactly one of the subsets (X is a disjoint union of the subsets)

Every Equivalence Relation on a set defines a partition of this set and every partition defines an equivalence relation.

Let $X = \{1, 2, 3, 4, 5, 6\}$ and relation R from X to X is defined as $x R y$, if and only if $x+y$ is greater than 8.

Find

1. R
2. R^{-1}
3. $\text{Dom}(R)$, $\text{Ran}(R)$
4. $\text{Dom}(R^{-1})$, $\text{Ran}(R^{-1})$

$$R = \{(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}$$

The binary relation

$R = \{(1, 1), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4)\}$ on a set $A = \{1, 2, 3, 4\}$ is

1. Reflexive, Symmetric and Transitive
2. Neither reflexive, nor Irreflexive but Transitive
3. Irreflexive, Symmetric and Transitive
4. Irreflexive and Anti-symmetric

Neither reflexive, nor irreflexive but transitive

$$R = \{(x, y) | x = y\}.$$

Identify if it is compatible, equivalence or partial order relation.

Compatible, Equivalence and Partial Order

Find the transitive closure of
 $R = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$

$$R = \{(1, 2), (2, 3), (3, 4), (4, 1)\}$$

$$R^2 = R \circ R = \{(1, 3), (2, 4), (3, 1), (4, 2)\}$$

$$R^3 = R^2 \circ R = \{(1, 4), (2, 1), (3, 2), (4, 3)\}$$

$$R^4 = R^3 \circ R = \{(1, 1), (2, 2), (3, 3), (4, 4)\}$$

$$R^\infty = R \cup R^2 \cup R^3 \cup R^4 =$$

$$\{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (3, 3), (3, 4), (4, 1), (4, 2), (4, 3), (4, 4)\}$$

Consider the relation R defined on $A = \{a, b, c, d\}$ as follows:

$$R = \{(a, b), (b, c), (d, c), (d, a), (a, d), (d, d)\}$$

Find the reflexive, transitive and symmetric closure of R

$$R^{(r)} = \{(a, a), (b, b), (c, c), (d, d), (b, c), (d, c), (d, a), (a, d)\}$$

$$R^{(s)} =$$

$$\{(a, b), (b, a), (b, c), (c, b), (d, c), (c, d), (d, a), (a, d), (d, d)\}$$

$$R^{\infty} =$$

$$\{(a, a), (a, b), (a, c), (a, d), (b, c), (d, a), (d, b), (d, c), (d, d)\}$$

Suppose $A = \{a, b, c, d\}$ and π_1 is the following partition of A

$$\pi_1 = \{\{a, b, c\}, \{d\}\}$$

1. List the ordered pair of equivalence relation induced by π_1 .
2. Draw graph of above equivalence relation.

$$R = \{(a,a), (a,b), (a,c), (b,a), (b,b), (b,c), (c,c), (c,b), (c,c), (d,d)\}$$